

MOHAMMED SHABAZ S

Software Developer | Embedded Engineer

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SUMMARY

Final-year Electronics & Communication Engineering student (CGPA 8.38) working across application software and embedded firmware. Internship experience at Nokia and iHelp Robotics, with projects in Android, computer vision, and microcontrollers using C/C++, Python, and Embedded C.

EDUCATION

Ballari Institute of Technology and Management (BITM) — Ballari, Karnataka

B.E. Electronics & Communication Engineering | Final Year | Expected 2027 | CGPA 8.38

Class 12: 80% | Class 10: 86.88%

SKILLS

Languages: C, C++, Python, Java, SQL

Software & Tools: Android Studio, OpenCV, Firebase, Git, GitHub

Embedded: Embedded C, Microcontrollers

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Nokia Solutions and Networks India — Student Intern

Bangalore | Sep 2026 – Present

- Student intern contributing to engineering work; project scope is covered by company confidentiality.

iHelp Robotics Private Limited — AI Research Intern, Deep Learning & Model Development

Remote | Mar 2026 – Aug 2026 | Project: MITRA

- Improved live camera-stream reliability in the MITRA assistive Android app: reduced WiFi/RTSP stream search delay, persisted the last working transport, and added refresh/stall recovery.
- Built a visual-freeze watchdog that reconnects when frame counters advance but the image stalls; added on-screen stream- and cloud-status panels and a guard that sends frames only when the cloud WebSocket is connected.
- Hardened offline voice commands (offline-preferred speech recognition with model recovery and microphone retry backoff); verified builds across devices with Gradle test/lint/assemble runs and captured logs. Backend and model work were handled by another team.

IEEE EMBS Pune Section — Student Intern, Skin Disease Classification

Remote | Jun 2026

- Built a demo-level skin-disease image classifier with a 3-person team: model selection, training, evaluation, image processing, UI, and testing in Python.

PROJECTS

VisionPay — Individual

Python, MobileNetV2, TensorFlow Lite, OpenCV | github.com/MdShabazS/visionpay

- Built an offline Indian-currency recognizer with spoken feedback for low-vision users: trained a MobileNetV2 model, converted it to TensorFlow Lite, and ran real-time webcam inference with text-to-speech.
- Collected ~400 images per denomination across 6 note classes; added confidence/margin gating, temporal smoothing, a background class, and auto-count mode; reported ~93% validation accuracy.

Automotive Body Control Module — Individual

ESP32, Embedded C | github.com/MdShabazS/Automotive-Body-Control-Module-ESP32

- Built an ESP32 body-control-module prototype with an OFF/ACC/ON ignition state machine, indicators, synchronized hazard mode, and an OLED dashboard.
- Structured the firmware around non-blocking millis()-based scheduling with brake debouncing, GPIO abstraction, and edge-triggered serial logging.

AEGIS — Team (lead), design stage / in progress

- Leading a 3-person team designing AEGIS, an AI-assisted emergency-response platform: defined a 15-stage incident workflow where AI analyzes and recommends and a human dispatcher verifies before resources are dispatched.

LEADERSHIP & ACTIVITIES

- Vice-Chair, IEEE CAS Society, IEEE Student Branch BITM (previously Treasurer)
- Treasurer, BITM Robotics Club
- Google Student Ambassador, 2025
- Selected Participant, IEEE SPACE 2026 — B.Tech Initiative

CERTIFICATIONS

Embedded Systems — Internshala Trainings (2025) | Python Programming — ElSystems (2024) | Google Cloud Generative AI — SmartBridge/SmartInternz (2025) | SQL for Data Analytics with AI | Programming in C and C++ with AI

LANGUAGES

English, Kannada, Hindi, Urdu